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Source: Sahil Bhola, Self-contained Proof for a Mixed-precision Block Jacobi Algorithm.

Selected deliverable: the full active result labeled theorem:solution_of_a_traingular_system in main.tex (lines
457-490).

The original mathematical proof is available in paper.tex. The selected statement below is authoritative.
Earlier statements and notation supply context. A referenced result may be proved as a helper. Do not assume a
desired algorithmic bound or an unproved paper lemma as an added hypothesis.

SELECTED STATEMENT (verbatim LaTeX, proof omitted)
\begin{theorem}[Solution of a Triangular System]\label{theorem:solution_of_a_traingular_system}
Let $\mat{t}\ireal{n\times n}$ be a nonsingular triangular matrix and let
$\vect{b}\ireal{n}$. Suppose that $\mat{t}\vect{x}=\vect{b}$ is solved by
forward substitution when $\mat{t}$ is lower triangular, or by backward
substitution when $\mat{t}$ is upper triangular. Assume that no overflow
or underflow occurs and that $n\urd<1$. Then the computed solution
$\hat{\vect{x}}$ satisfies
\begin{align}
    (\mat{t}+\Delta\mat{t})\hat{\vect{x}}=\vect{b},
    \qquad
    \abs{\Delta\mat{t}}\leq\gamma_n\abs{\mat{t}},
\end{align}
where the inequality is interpreted componentwise,
$\Delta\mat{t}$ has the same triangular structure as $\mat{t}$, and
$\gamma_n=n\urd/(1-n\urd)$.
\end{theorem}

SECTION CONTEXT
\subsection{Triangular Solve}
For solving triangular systems, a key computational kernel that arises is
\begin{align}\label{eq:traingular_kernel}
    y = \frac{c - \sum_{i=1}^{k-1}\vect{a}_i\vect{b}_i}{\vect{b}_k},
\end{align}
where $\vect{a},\vect{b}\ireal{k}$.
In the following lemma, we first perform a backward error analysis for~\cref{eq:traingular_kernel}.

[Earlier statement and proof omitted here.]


Now, using~\cref{lemma:traingular_update} we can perform a backward error analysis for solving a traingular
system $\mat{t}\vect{x}=\vect{b}$ for $\mat{t}\ireal{n\times n}$ in floating-point arithmetic, as stated below.

EARLIER STATEMENTS FOR CONTEXT
\begin{lemma}[Traingular Update]\label{lemma:traingular_update}
Let $k\geq2$, $c\in\R$, and $\vect{a},\vect{b}\ireal{k}$ with
$\vect{b}_k\neq0$. Suppose that
\begin{align}
    y=\frac{c-\sum_{i=1}^{k-1}\vect{a}_i\vect{b}_i}{\vect{b}_k}
\end{align}
is evaluated in floating-point arithmetic by computing the inner product
sequentially, followed by the subtraction and division. Assume that no
overflow or underflow occurs and that $k\urd<1$. Then the computed result
$\hat{y}$ satisfies the perturbed equation
\begin{align}
    \sum_{i=1}^{k-1}
        \vect{a}_i\left(\vect{b}_i+\Delta\vect{b}_i\right)
    +\hat{y}\left(\vect{b}_k+\Delta\vect{b}_k\right)
    =c,
\end{align}
where
\begin{align}
    \abs{\Delta\vect{b}_i}
    \leq \gamma_k\abs{\vect{b}_i},
    \qquad i=1,\ldots,k,
\end{align}
and $\gamma_k=k\urd/(1-k\urd)$.
\end{lemma}

COMMON ROUNDING MODEL AND ERROR CONSTANTS
\section{Floating-Point Model}
We assume the standard floating-point model~\cite{higham2002accuracy}: for every elementary
operation $\circ \in \{+,-,\times,/\}$,
\begin{equation}
    \fl(a \circ b) = (a \circ b)(1+\delta),
    \qquad
    \abs{\delta} \le \urd,
    \label{eq:fp-model}
\end{equation}
where $\urd$ is the unit roundoff and $\delta$ is the relative roundoff error.
Using~\cref{eq:fp-model} for numerical analysis we typicall obtain the product
\begin{align}
    \prod_{i=1}^n (1 + \delta_i)^{\rho_i} \define 1 + \theta_n, \qquad \rho = \pm 1,
    \label{eq:theta_n}
\end{align}
where $n$ denotes the number of accumulated roundoff error terms.
For numerical analysis, we seek reliable and sharp bounds for $\theta_n$.
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\section{Deterministic Rounding Error Analysis}
The deterministic rounding error analysis proposes the following problem-size dependent constant
\begin{align}
\gamma_{n} \define \frac{n\urd}{1 - n\urd},
\label{eq:drea}
\end{align}
for $n\urd < 1$.
Here, $n$ is the number of accumulated error terms~\cite{higham2002accuracy}.
The problem-size dependent constant is monotonic, that is, $\gamma_{n+1} > \gamma_{n}$ for $\urd > 0$.
Performing backward/forward error analysis using~\cref{eq:drea} we obtain
\begin{align}
    \abs{\theta_n} \leq \gamma_n.
\end{align}

SOURCE NOTATION MACROS
\newcommand{\urd}{\ensuremath{\mathrm{u}}}
\newcommand{\define}{\ensuremath{\triangleq}}
\newcommand{\vect}[1]{\mathbf{\MakeLowercase{#1}}}
\newcommand{\mat}[1]{\mathbf{\MakeUppercase{#1}}}
\newcommand{\ireal}[1]{\in\mathbb{R}^{#1}}
\newcommand{\norm}[2][]{\left\|#2\right\|_{#1}}
\DeclareMathOperator{\fl}{fl}
\newcommand{\R}{\mathbb{R}}
\newcommand{\abs}[1]{\left|#1\right|}
\newcommand{\ceil}[1]{\left\lceil #1 \right\rceil}
\newcommand{\widehatgamma}[1]{\widehat{\gamma}_{#1}}
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